1 Recurrences Substitution

Problem 1.1. Express the **worst case running time** of each of the following functions as a recurrence relation. Be sure to include a base case.

YOU DO NOT NEED TO ANALYZE THE RECURRENCE RELATION

(a)

```
function Func(A[],n)
    if n ≤ 6 then
        return A[1]
    end if
    x = 0
    for i = 1 to n do
        for j = 1 to n do
            x = x + i − j
        end for
    end for
    x = x + Func(A[], n − 10)
    return x
end function
```

$T(n) = cn^2 + T(n-10)$

$T(6) = c$

(b)

```
function Func(A[],n)
    if n ≤ 64 then
        return A[1]
    end if
    x = 0
    for i = 1 to n do
        for j = 1 to n do
            x = x + i − j
        end for
    end for
    if A[1] ≤ A[n] then
        x = x + Func(A[], ⌊n/6⌋)
    else
        x = x + Func(A[], n − 6)
    end if
    return x
end function
```

$T(n) = cn^2 + T(n-6)$

$T(64) = c$

(c) Let $n = j - i + 1$

```
function Func(A[],i,j)
    if i < j then
        x = 0
        for m = i to j do
            x = x + A[i] - A[j]
        end for
        if A[i]^2 ≤ A[j] then
            x = x + Func(A[], i + 2, j)
        else
            x = x + Func(A[], i, j - 3)
        end if
        return x
    else
        return 0
    end if
end function
```

$c(j-i+1) = cn$

$s_1 = j - (i+2) + 1 = n - 2$

$s_2 = j - 3 - i + 1 = n - 3$

$T(n) = cn + T(n-2)$

$T(1) = c$

Problem 1.2. Express the **best case running time** of each of the following functions as a recurrence relation. Be sure to include a base case.

YOU DO NOT NEED TO ANALYZE THE RECURRENCE RELATION

(a)

```
function Func(A[],n)
    if n ≤ 4 then
        return A[1]
    end if
    x = 0
    for i = 1 to n do
        for j = 1 to n do
            x = x + i − j
        end for
    end for
    x = x + Func(A[], n − 10)
    return x
end function
```

$T(n) = cn^2 + T(n-10)$

$T(4) = c$

(b)

```
function Func(A[],n)
    if n ≤ 64 then
        return A[1]
    end if
    x = 0
    for i = 1 to n do
        for j = 1 to n do
            x = x + i − j
        end for
    end for
    if A[1] ≤ A[n] then
        x = x + Func(A[], ⌊n/6⌋)
    else
        x = x + Func(A[], n − 6)
    end if
    return x
end function
```

$T(n) = cn^2 + T(n/6)$

$T(64) = c$

Problem 1.3. Determine the asymptotic complexity of each of the following recurrence relations. Justify your answer with either substitution, a recursion tree, or induction. If the problem takes at least exponential time, then simply find exponential lower bounds.

(a)

$$T(n) = cn^2 + T(n-15), \qquad T(0) = c$$

$$T(n) = cn^2 + c(n-15)^2 + T(n-30)$$

$$T(n) = cn^2 + c(n-15)^2 + c(n-30)^2 + T(n-45)$$

For any value of $k \geq 1$,

$$T(n) = \sum_{i=0}^{k-1} c(n-15i)^2 + T(n-15k)$$

Choose k s.t.

$n - 15k = 0$

$k = n/15$

$$T(n) = \sum_{i=0}^{\frac{n}{15}-1} c(n-15i)^2 + \underbrace{T(0)}_{c}$$

UB:

$$T(n) \leq \sum_{i=0}^{\frac{n}{15}-1} cn^2 + c$$

$$T(n) \leq \frac{n}{15} \cdot cn^2 + c$$

$O(n^3)$

LB:

$$T(n) \geq \sum_{i=0}^{n/30} c(n-15i)^2$$

$$T(n) \geq \sum_{i=0}^{n/30} c\left(n - 15 \cdot \frac{n}{30}\right)^2$$

$$T(n) \geq \left(\frac{n}{30}+1\right) c\left(n - \frac{n}{2}\right)^2$$

$\Omega(n^3)$

Thus, $T(n) \in \Theta(n^3)$.

(b)

$$T(n) = c\sqrt{n} + T(n/4), \qquad T(5) = c$$

$$T(n) = c\sqrt{n} + c\sqrt{\frac{n}{4}} + T(n/16)$$

$$T(n) = c\sqrt{n} + c\sqrt{\frac{n}{4}} + c\sqrt{\frac{n}{16}} + T(n/64)$$

For any value of $k \geq 1$,

$$T(n) = \sum_{i=0}^{k-1} c\sqrt{\frac{n}{4^i}} + T(n/4^k)$$

$$T(n) = \sum_{i=0}^{k-1} \frac{c\sqrt{n}}{2^i} + T(5)$$

$$T(n) = c\sqrt{n} \sum_{i=0}^{k-1} (1/2)^i + c$$

Choose k s.t.

$$\frac{n}{4^k} = 5$$

$$k = \log_4(n/5)$$

UB: $T(n) \leq c\sqrt{n} \sum_{i=0}^{\infty} (1/2)^i + c$

$T(n) \leq c\sqrt{n} \left(\frac{1}{1-1/2}\right) + c$

LB: $T(n) \geq c\sqrt{n} \cdot (1/2)^0$

Thus, $T(n) \in \Theta(\sqrt{n})$

(c)

$$T(n) = c + T(\sqrt{n}), \qquad T(2) = c$$

Hint: This problem is weird, but surprisingly easy.

$T(n) = c + c + T(n^{1/4})$

$T(n) = c + c + c + T(n^{1/8})$

In general,

$T(n) = kc + T(n^{1/2^k})$

$T(n) = \log_2(\log_2(n)) \cdot c + T(5)$

Thus, $T(n) \in \Theta(\log(\log(n)))$

Choose k s.t.

$n^{1/2^k} = 2$

$n = 2^{2^k}$

$\log_2(n) = 2^k$

$k = \log_2(\log_2(n))$

2 Multiple Recurrences

Problem 2.1. Express the **worst case running time** of each of the following functions as a recurrence relation. Be sure to include a base case.

(a)

```
function Func(A[],n)
    if n ≤ 16 then
        return A[1]
    end if
    x = 0
    for i = 1 to 8 do
        for j = 1 to n do
            x = x + i − j
        end for
        x = x + Func(A[], ⌊n/5⌋)
    end for
    return x
end function
```

$T(n) = cn + 8T(n/5)$

$T(16) = c$

(b)

```
function Func(A[],n)
    if n ≤ 36 then
        return A[1]
    end if
    x = 0
    for i = 1 to 3 do
        for j = 1 to n do
            x = x + i − j
        end for
        x = x + Func(A[], ⌊√n⌋)
    end for
    if A[1] ≤ A[n] then
        x = x + Func(A[], ⌊n/4⌋)
    else
        x = x + Func(A[], n − 4)
    end if
    return x
end function
```

$T(n) = cn + 3T(\sqrt{n}) + T(n-4)$

$T(36) = c$

Problem 2.2. Determine the asymptotic complexity of each of the following recurrence relations. Justify your answer with either substitution, a recursion tree, or induction. If the problem takes at least exponential time, then simply find exponential lower bounds.

(a)

$$T(n) = cn^2 + 12T(n/3), \qquad T(1) = c$$

$$T(n) = cn^2 + 12\left[c(n/3)^2 + 12T(n/9)\right]$$

$$T(n) = cn^2 + \frac{12cn^2}{9} + 12^2 T(n/9)$$

$$T(n) = cn^2 + \frac{12cn^2}{9} + 12^2\left[c(n/9)^2 + 12T(n/27)\right]$$

$$T(n) = cn^2 + \frac{12cn^2}{9} + \frac{12^2cn^2}{9^2} + 12^3 T(n/27)$$

In general,

$$T(n) = \sum_{i=0}^{k-1} \left(\frac{12}{9}\right)^i cn^2 + 12^k T(n/3^k)$$

Choose k s.t.

$$\frac{n}{3^k} = 1$$

$$k = \log_3(n)$$

$$T(n) = cn^2 \sum_{i=0}^{k-1} \left(\frac{4}{3}\right)^i + 12^{\log_3(n)} T(1)$$

$$T(n) = cn^2\left[\frac{(4/3)^k - 1}{4/3 - 1}\right] + n^{\log_3(12)} \cdot c$$

$$T(n) = 3cn^2\left[(4/3)^{\log_3(n)} - 1\right] + cn^{\log_3(12)}$$

$$T(n) = \underbrace{3cn^2\left[n^{\log_3(4/3)} - 1\right]}_{\approx n^{2+\log_3(4)-\log_3(3)}} + cn^{\log_3(12)}$$

$$= n^{1+\log_3(4)}$$

$$= n^{\log_3(3)+\log_3(4)}$$

$$= n^{\log_3(12)}$$

Thus, $T(n) \in \Theta(n^{\log_3(12)})$

(b)

$$T(n) = cn + \overbrace{T(n-2) + T(n-6) + T(n-3)}^{\text{more than one chip} \;=\; \text{EXP. LB.}}, \qquad T(5) = c$$

$T(n) \geq T(n-6) + T(n-6) + T(n-6)$

$T(n) \geq 3T(n-6)$

$T(n) \geq 3 \cdot 3T(n-12)$

$T(n) \geq 3 \cdot 3 \cdot 3T(n-18)$

In general,

$T(n) \geq 3^k T(n-6k)$

$T(n) \geq 3^{\frac{n-5}{6}} T(5)$

Choose k s.t.

$n - 6k = 5$

$k = \frac{n-5}{6}$

Thus, $T(n) \in \Omega(3^{n/6})$

exp. lower bound!

(c)

$$T(n) = cn + \alpha T(n/\beta), \qquad T(1) = c$$

Assume that $\alpha, \beta > 0$ and$\alpha < \beta$. This should be "no harder" than if α, β were numbers!

$$T(n) = cn + \alpha\left[\frac{cn}{\beta} + \alpha T(n/\beta^2)\right],$$

$$T(n) = cn + c\left(\frac{\alpha}{\beta}\right)n + \alpha^2 T(n/\beta^2)$$

$$T(n) = cn + c\left(\frac{\alpha}{\beta}\right)n + \alpha^2\left[\frac{cn}{\beta^2} + \alpha T(n/\beta^3)\right]$$

$$T(n) = cn + c\left(\frac{\alpha}{\beta}\right)n + c\left(\frac{\alpha}{\beta}\right)^2 n + \alpha^3 T(n/\beta^3)$$

In general,

$$T(n) = \sum_{i=0}^{k-1} cn\left(\frac{\alpha}{\beta}\right)^i + \alpha^k T(n/\beta^k)$$

$$T(n) = cn \sum_{i=0}^{k-1} (\alpha/\beta)^i + \alpha^{\log_\beta(n)} \cdot c$$

Choose k s.t.

$$\frac{n}{\beta^k} = 1$$

$$k = \log_\beta(n)$$

1) Since $\alpha < \beta$, $\frac{\alpha}{\beta} < 1$.

2) Since $\alpha < \beta$,

$$\log_\beta(\alpha) < \log_\beta(\beta) = 1.$$

UB: $T(n) \le cn \sum_{i=0}^{\infty} (\alpha/\beta)^i + n^{\log_\beta(\alpha)} \cdot c$

$T(n) \le cn\left(\frac{1}{1-\alpha/\beta}\right) + cn^{\log_\beta(\alpha)}$ $\Big\}$ $O(n)$

LB: $T(n) \ge cn(\alpha/\beta)^0 = cn$ $\Big\}$ $\Omega(n)$

Thus, $T(n) \in \Theta(n)$.

3 Recursion Trees

Problem 3.1. Express the **worst case running time** of each of the following functions as a recurrence relation. Be sure to include a base case.

(a)

```
function Func(A[],n)
    if n ≤ 16 then
        return A[1]
    end if
    x = 0
    for i = 1 to n do
        for j = 1 to n do
            x = x + i − j
        end for
    end for
    x = x + Func(A[], ⌊n/6⌋)
    x = x + Func(A[], ⌊n/1.3⌋)
    return x
end function
```

$T(n) = cn^2 + T(n/6) + T(n/1.3)$

$T(16) = c$

(b)

```
function Func(A[],n)
    if n ≤ 36 then
        return A[1]
    end if
    x = 0
    for i = 1 to 3 do
        for j = 1 to n do
            x = x + i − j
        end for
        x = x + Func(A[], ⌊√n⌋)
    end for
    x = x + Func(A[], ⌊n/2⌋)
    return x
end function
```

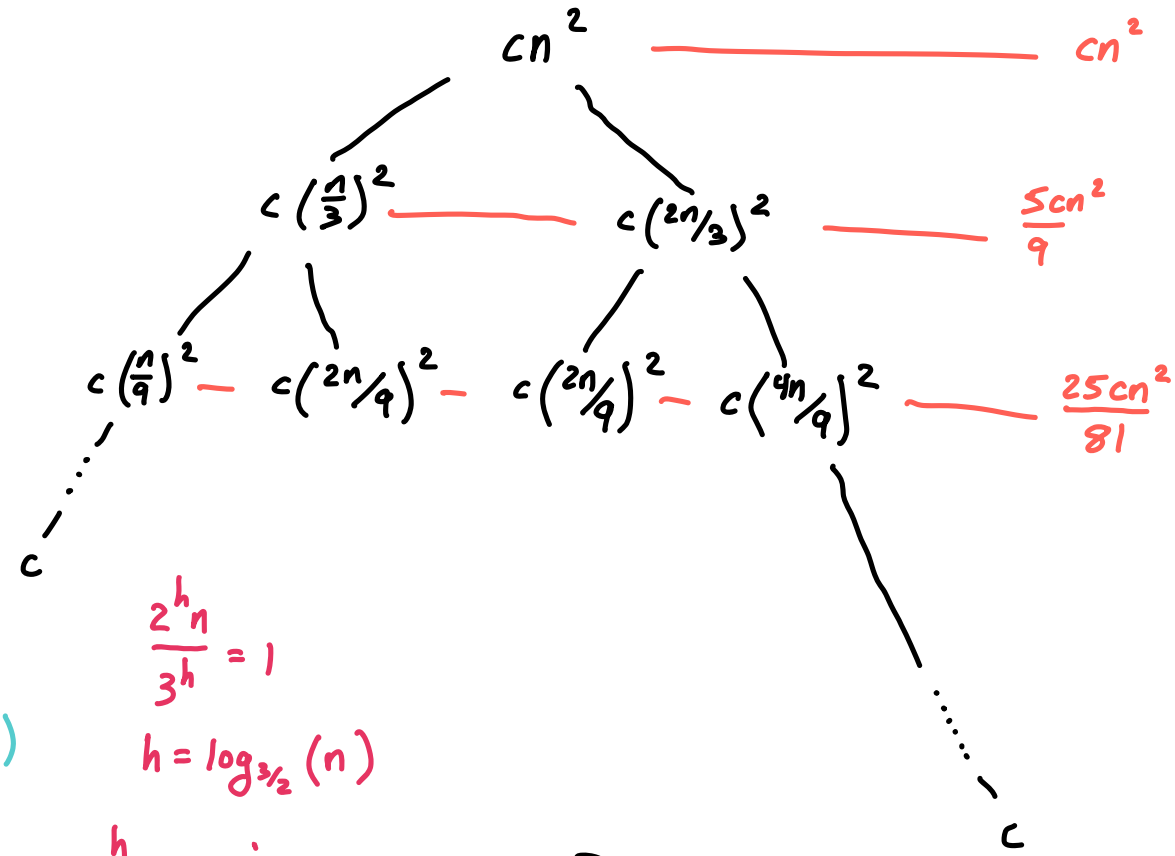
$T(n) = cn + 3T(\sqrt{n}) + T(n/2)$

$T(36) = c$

Problem 3.2. Determine the asymptotic complexity of each of the following recurrence relations. Justify your answer with either substitution, a recursion tree, or induction. If the problem takes at least exponential time, then simply find exponential lower bounds.

(a)

$$T(n) = cn^2 + T(n/3) + T(2n/3), \qquad T(1) = c$$



$\frac{n}{3^k} = 1$

$k = \log_3(n)$

$\frac{2^h n}{3^h} = 1$

$h = \log_{3/2}(n)$

$$T(n) \le \sum_{i=0}^{h} \left(5/9\right)^i cn^2$$

$$T(n) \le cn^2 \sum_{i=0}^{\infty} \left(5/9\right)^i = cn^2\left[\frac{1}{1-5/9}\right]$$

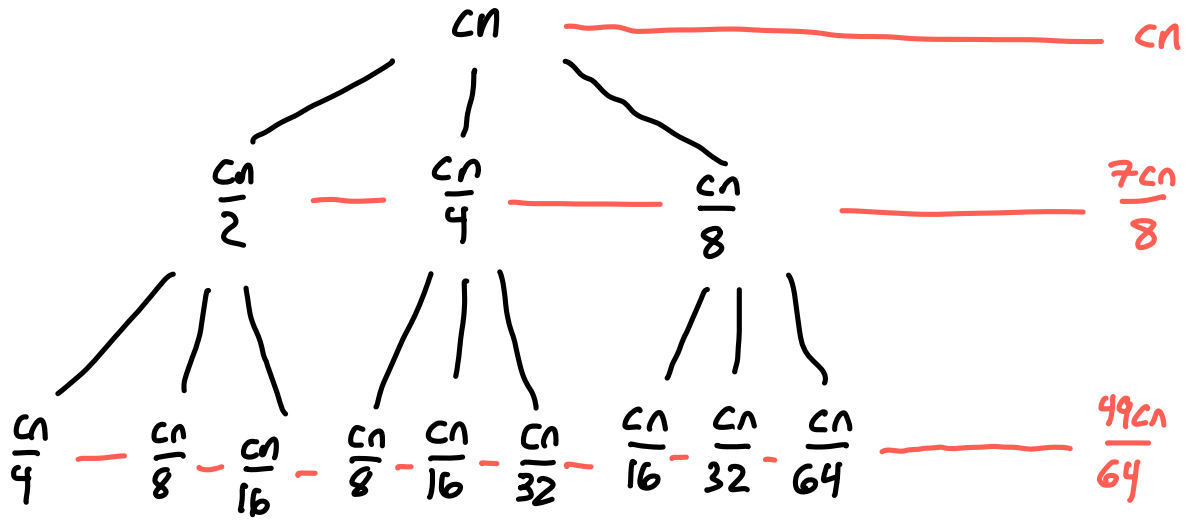
$$T(n) \ge \sum_{i=0}^{k} \left(5/9\right)^i cn^2$$

$$T(n) \ge cn^2$$

Thus, $T(n) \in \Theta(n^2)$.

(b)

$$T(n) = cn + T(n/2) + T(n/4) + T(n/8), \qquad T(5) = c$$



$\frac{n}{2^k} = 1$

$k = \log_2(n)$

$\frac{n}{8^h} = 1$

$h = \log_8(n)$

$$T(n) \le \sum_{i=0}^{h} \left(\tfrac{7}{8}\right)^i cn$$

$$T(n) \le cn \sum_{i=0}^{\infty} \left(\tfrac{7}{8}\right)^i = cn \left[\frac{1}{1-7/8}\right]$$

$$T(n) \ge \sum_{i=0}^{k} \left(\tfrac{7}{8}\right)^i cn$$

$$T(n) \ge cn$$

$T(n) \in \Theta(n)$.

Problem 3.3. Determine the asymptotic complexity of the following recurrence relation. Justify your answer with **A RECURSION TREE**.

Note: the answer here will look silly.

typo.

$$T(n) = cn^3 + T(n/2), \qquad T(0) = c$$

$$\begin{array}{c} cn^3 \\ | \\ c\left(\frac{n}{2}\right)^3 \\ | \\ c\left(\frac{n}{4}\right)^3 \\ \vdots \\ c \end{array}$$

$\frac{n}{2^h} = 1$

$h = \log_2(n)$

$$T(n) = \sum_{i=0}^{h} c\left(n/2^i\right)^3 = cn^3 \sum_{i=0}^{h} (1/8)^i$$

$$T(n) \le cn^3 \sum_{i=0}^{\infty} (1/8)^i = cn^3\left(\frac{1}{1-1/8}\right)$$

$$T(n) \ge cn^3$$

Thus, $T(n) \in \Theta(n^3)$.